

Chapter 21 — Urologic Considerations in Pregnancy

PHYSIOLOGIC CHANGES DURING PREGNANCY

- Pregnancy adapts **almost every organ system**; knowing normal adaptation lets the urologist **separate physiology from pathology**.

Cardiovascular

- **Fall in systemic vascular resistance (SVR) is the primary driver**: low-resistance placenta plus progesterone- and prostacyclin-mediated vasodilatation, evident **before** uteroplacental circulation exists.
- **Mean arterial pressure falls ~10 mmHg below baseline by the second trimester**.
- Sympathetic drive → **heart rate rises up to 30%, maximal at 20 weeks**.
- **Cardiac output (CO) = stroke volume × heart rate**; both rise → CO rises early and **peaks 30–50% above preconception values**.
- **CO is position-dependent**: supine CO at term **up to 30% below lateral recumbency**; reduced venous return → stroke volume falls, possibly **abruptly**.
- Uterus compresses the **inferior vena cava (IVC) ± aorta** → **supine hypotensive syndrome**, worsening with gestation; late supine lower-limb return may depend **entirely on collaterals**.
- **General and regional anaesthesia abolish sympathetic compensation** (SVR cannot rise) → further CO fall and hypotension; **treat maternal hypotension aggressively** to protect uteroplacental perfusion.
- Prevention: **avoid supine and lithotomy**; **right-side elevation (15–20° left tilt)** displaces the uterus off the IVC.
- Venous compression also → oedema, varicosities, haemorrhoids and **deep venous thrombosis (DVT) risk**.

Concept — *Falling resistance with rising output defines the pregnant circulation; the gravid uterus turns supine positioning into a preload problem and anaesthesia removes the compensation — hence left tilt for every urological procedure.*

Respiratory

- Ligamentous relaxation → **chest circumference increases up to 7 cm**.
- Diaphragmatic excursion **preserved**, but **diaphragm displaced cranially up to 4 cm** → lung volumes fall despite chest-wall expansion.
- Mechanical restriction: **total lung capacity, expiratory reserve volume and residual volume fall**; **functional residual capacity (FRC) falls to ~80% of baseline by term**.
- **Progesterone drives chronic hyperventilation**: tidal volume up to **+50% by 8 weeks** → minute ventilation ↑ → alveolar O_2 ↑, CO_2 ↓ → facilitates **fetal-to-maternal CO_2 transfer**.
- Low maternal carbon dioxide = **chronic respiratory alkalosis**, partially compensated by **renal bicarbonate excretion**.
- **Oxygen consumption up to +40% + reduced FRC** → **small oxygen reserve** → **rapid desaturation at intubation**.
- **Difficult airway**: oropharyngeal oedema, narrower glottic aperture, mucosal friability → higher airway classification; **obesity and pre-eclampsia** compound it; supine positioning further lowers FRC under general anaesthesia.

Concept — *Pregnancy raises oxygen demand while shrinking the oxygen store (FRC); the anaesthetic danger is fast desaturation in a harder airway.*

Haematologic

- **Blood volume expands 40–50% from ~6 weeks; plasma volume exceeds red blood cell (RBC) mass** → haematocrit falls: **dilutional (physiological) anaemia**, nadir 30–34 weeks, partly ameliorated by iron.
- **Mild gestational thrombocytopenia; a platelet count below 116,000/ μ L is exceedingly rare and mandates evaluation**, especially perioperatively. **Platelet function and haemostasis are intact.**
- **Hypercoagulability** guards against placental-bed haemorrhage; **venous thromboembolism (VTE) risk ~10-fold** via **Virchow triad**: stasis (IVC compression), vascular trauma, $\uparrow$ procoagulants.
- Pregnancy-independent VTE risk factors: **age >35**, obesity, trauma, immobility, infection, smoking, nephrotic syndrome, coagulopathy, cancer, orthopaedic surgery, **previous VTE**.
- VTE is uncommon but **a leading cause of maternal death (~9% of pregnancy-related deaths)**; pulmonary embolism arises predominantly from lower-limb DVT.
- **Hospitalisation independently raises VTE risk ~17-fold**; surgery adds arrhythmia, hypoxia, pulmonary hypertension and heart failure.
- Prophylaxis (extrapolated from caesarean data): **compression stockings and pneumatic compression** — **favourable risk–benefit**; **low-dose unfractionated heparin (UFH)** for select high risk, with obstetrics and anaesthesia.
- **Therapeutic anticoagulation is contraindicated with most regional anaesthesia**; UFH is reversible with **intravenous protamine sulfate**. **Left lateral decubitus** reduces IVC compression and postoperative DVT.
- **Physiological leucocytosis**: second/third trimester mean white blood cell (WBC) count **8500/ mm^3 (range 5600–12,200)**, **peaking up to 30,000/ mm^3 in labour** → **WBC is unreliable for diagnosing infection**.

Concept — *Pregnancy haematology is dilution plus hypercoagulability: low haematocrit and platelets are expected, thrombosis is the real threat, and the WBC count loses its value as an infection marker.*

Renal and Urinary Tracts

- Volume expansion and vasodilatation → $\uparrow$ renal blood flow → **glomerular filtration rate (GFR) +40–65%**.
- Serum creatinine, urea and uric acid fall: **mean creatinine ~0.8 mg/dL; blood urea nitrogen (BUN) 13** → **8–10 mg/dL**.
- Tubular reabsorption balances sodium, but the **osmotic threshold for antidiuretic hormone (ADH) release and thirst is reset lower** → **mild hyponatraemia, sodium 4–5 mmol/L below baseline**.
- **Kidneys enlarge up to 1 cm**, partly from dilatation of the whole collecting system.
- **Pelvicalyceal dilatation**: historically up to **90% of pregnancies**, reaching **15 mm on the right and 5 mm on the left** by term; contemporary sonographic reference data show renal pelvis dilatation **>10 mm is rare**.
- **Dilatation begins very early** → **not purely mechanical**: progesterone, relaxin and endothelin relax smooth muscle and increase **urinary dead space**; it occurs even in **ectopic kidneys (pelvic transplant)**.
- **Mechanical compression by the gravid uterus, predominantly right-sided**; dilatation starts by the second month, **peaks in the second trimester**, ureteric diameter up to **2 cm**.
- Obstruction becomes harder to discern; **pyelonephritis risk rises with pre-existing bacteriuria**.
- **Complete postpartum resolution** expected, barring intra-abdominal pathology.

Urine Chemistries During Pregnancy

- **Glycosuria is very common (10–100-fold increase) but intermittent**; repeated glycosuria → obstetric screening for diabetes.
- Hyperfiltration $\pm$ reduced tubular reabsorption raise protein excretion: **upper limits per 24 h** — **proteinuria 300 mg, albuminuria 30 mg**; rise is steepest in first and second trimesters.

- **Dipstick proteinuria is inaccurate** (ignores urinary concentration); **persistent proteinuria** → exclude diabetes, pre-eclampsia, renal disease, urinary tract infection (UTI).
- Fetal calcium demand + low albumin → **total calcium falls**; **ionised calcium is unchanged**; **calciuria doubles**.

Hematuria in the Pregnant Patient

- Uterine compression **elevates the trigone and increases bladder vascularity** → **idiopathic microscopic haematuria on dipstick in up to 20%**.
- Usually benign; commonest causes: **infection, urolithiasis, renal disease, medications, trauma, obstruction, tumours**.
- **Pregnancy-exclusive causes: pre-eclampsia and placenta percreta**.
- **Pre-eclampsia** = systemic syndrome of **new-onset hypertension and proteinuria after 20 weeks**; microscopic haematuria at **two to eight times normal**.
- Most literature suggests an association between idiopathic haematuria and later pre-eclampsia, but larger cohorts found **no association with complications** → **value of haematuria screening remains unclear**.
- **Trap**: a nonpregnant-normal creatinine may represent impairment, and a raised WBC is not evidence of infection.

Concept — *Hydronephrosis of pregnancy is hormonal early and mechanical later, right-predominant and self-resolving, so dilatation alone never establishes obstruction; rising GFR likewise lowers every filtration marker, so “normal” values shift downward.*

PHYSIOLOGIC HYDRONEPHROSIS OF PREGNANCY

- The core consult: **physiological versus pathological dilatation**, especially with colic or suspected anatomical obstruction. **Structure and function do not correlate**; **pathological hydronephrosis is rare** (symptomatic admissions a tiny fraction of deliveries; physiological hydronephrosis common but symptomatic in only a small minority).
- **Flank pain correlates poorly with hydronephrosis**, and **grade does not correlate with pain intensity**; grade does not affect birth weight.
- **Spontaneous renal fornix rupture (SRFR)**: rare (first described 1947), usually of unknown cause beyond pregnancy; **commonest in first pregnancy and third trimester**; **pain is the only consistent symptom**, laboratory findings variable, **diagnosed on imaging**; no long-term urological sequelae, but **significantly more preterm labour**.

Management of symptomatic physiological hydronephrosis (no calculus)

- **Conservative first: analgesia, positioning, intravenous fluids**. Failure = **evolving infection or deteriorating renal function**.
- Randomised stent-versus-conservative trial: stenting gave better relief in moderate-severe hydronephrosis, **offset by stent discomfort**, and few conservative patients crossed over → **favour conservative care**.
- Double-J stenting under local anaesthesia associated with **more caesarean section, preterm labour, premature rupture of membranes, lower Apgar scores, postpartum UTI**.
- Where offered a stent, over half chose conservative care with **no difference in complications, postpartum pain or persistent hydronephrosis** → **stents reserved for complicated cases or additional obstructing pathology**; intervention otherwise needed mainly for overt pyelonephritis.
- **Predictors of eventual stenting**: maximal anteroposterior renal pelvis diameter, parity, C-reactive protein (CRP), WBC count, pain level.

- Stent retrospective series are **contaminated by stone indications**. Risks: flank pain, irritative voiding symptoms, **rapid encrustation (hypercalciuria, hyperuricosuria)** → individualise exchange (intervals tabulated under Urolithiasis).
- **Percutaneous nephrostomy (PCN)** for refractory cases, **very selectively** absent obstruction (duration, encrustation, discomfort).

PHYSIOLOGICAL HYDRONEPHROSIS VERSUS OBSTRUCTING CALCULUS

	Physiological hydronephrosis	Obstructing calculus
Side	Right-predominant	Left = right
Level	Stops at pelvic brim; rare below	Anywhere, including distal ureter
Onset	From 6–10 weeks, peaks second trimester	Presents second/third trimester
Resistive index	Unchanged	Raised
First-line care	Analgesia, position, fluids	Hydration, analgesia, trial of passage
Drainage	Only for infection / failing function	Urgent if septic; ureteroscopy if conservative care fails

Concept — *Because dilatation is expected and correlates poorly with symptoms, the default is conservative; drainage is earned by infection, failing function or a genuine obstructing cause.*

GENERAL CONSIDERATIONS FOR IMAGING OF THE PREGNANT PATIENT

- Fetal radiation effects: **teratogenesis, carcinogenesis, mutagenesis**; risk depends on **gestational age and dose**.
- **First trimester (organogenesis, rapid division) = highest teratogenic sensitivity**; second/third trimester exposure is less teratogenic but **raises childhood malignancy risk**.
- **American College of Obstetricians and Gynecologists (ACOG)** guidance (2017): individualise; **most urological imaging doses are below the threshold for fetal harm**; plan with the radiologist to lower dose or acoustic output.

ACOG practice recommendations for diagnostic imaging

- **Ultrasonography (US) and magnetic resonance imaging (MRI)** carry no associated risk and are the **techniques of choice** — used prudently, only to answer a relevant clinical question.
- With few exceptions, **radiography, computed tomography (CT) and nuclear medicine deliver doses far below those associated with fetal harm**; if necessary or more readily available, **they should not be withheld**.
- **Gadolinium limited**: only if it significantly improves diagnostic performance and is expected to improve maternal or fetal outcome.
- **Nuclear medicine selectively**, depending on the radioisotope. **Technetium-99m mercaptoacetyltriglycine (^{99m}Tc-MAG3)** — most widely used renal agent; **not filtered, extracted by tubules**; differential function and drainage. **^{99m}Tc studies give fetal exposure <5 mGy (safe); half-life 6 h, pure gamma emitter.**
- **Intraoperative fluoroscopy**: shield the fetus, **collimate, low-dose spot films** at the lowest feasible exposure for retrograde pyelography and stent positioning.

Concept — *Fetal risk is the product of gestational age and dose; because diagnostic doses sit below the harm threshold, necessary imaging is minimised, never withheld.*

GENERAL CONCEPTS FOR SURGICAL MANAGEMENT OF THE PREGNANT PATIENT

- ACOG guidance on nonobstetric surgery (updated 2019); **obstetric consultation always**, including the need for fetal monitoring.

ACOG practice recommendations for nonobstetric surgery

- **No current anaesthetic agent is teratogenic in humans** at standard concentrations at any gestational age.
- **No evidence of fetal brain effect** from in-utero anaesthetic or sedative exposure; no animal data for exposures <3 hours.
- **Fetal heart rate (FHR) monitoring** can guide positioning and cardiorespiratory management and may influence a decision to deliver.
- **Never deny or delay medically necessary surgery, regardless of trimester.**
- **Postpone elective surgery until after delivery.**
- **Antenatal corticosteroids** for fetuses at viable preterm gestations; monitor for preterm labour perioperatively.
- **Screen for VTE risk and give perioperative prophylaxis.**
- Operate where there are **neonatal and paediatric services**, an obstetric provider with **caesarean privileges** readily available, and someone qualified to **interpret FHR**.
- Ethics: involve all stakeholders, but **patient autonomy comes first** (ACOG ethics committee opinion); counsel risks to mother and fetus honestly; **women are not denied procedures because they are pregnant.**
- **Left lateral decubitus** (modified) for open, laparoscopic and endoscopic procedures.
- Approach by surgeon experience and exposure; **no randomised trials; open and laparoscopic outcomes broadly equal** (laparoscopy associated with earlier delivery and lower birth weight in one series).
- **Society of American Gastrointestinal and Endoscopic Surgeons (SAGES):** ports placed under direct vision according to **fundal height; insufflation 10–15 mmHg is safe**; higher pressures cause maternal cardiorespiratory change **after 20 minutes**; use **maternal end-tidal capnography**.
- **Excess perinatal mortality is due to the disease, not the operation.** Third-trimester surgery is associated with **preterm delivery**, and **third-trimester ureteroscopy with preterm labour** — key for counselling.
- ACOG now permits necessary surgery in all trimesters, but the **second trimester carries the least risk of preterm contractions and spontaneous abortion.**

Concept — *Necessity, not trimester, decides whether to operate; the second trimester is the preferred window, and untreated disease — not surgery — is what drives perinatal loss.*

GENERAL CONCEPTS OF UROLOGIC MEDICATION ADMINISTRATION DURING PREGNANCY

- Prescription drug use in pregnancy is **almost universal**.
- Physiological change alters **pharmacokinetics** — **absorption, distribution, metabolism, excretion**.
- Placental transfer favours **uncharged, lipophilic, low-molecular-weight** drugs.
- **US Food and Drug Administration (FDA) Pregnancy and Lactation Labeling Rule (PLLR), 2015:** letter categories **A, B, C, D and X removed** from all prescription drug and biological labelling.
- Label must now carry: **a risk summary for pregnancy and lactation, the supporting data, and information to guide prescribing and counselling.** Consequence: **individualised evidence review**; rarely a clear “yes” or “no”.
- Use the FDA searchable database and consult the patient's obstetrician.
- **OnabotulinumtoxinA (BoNT-A)** (idiopathic overactive bladder, neurogenic lower urinary tract symptoms): manufacturer states **may cause fetal harm** on animal data — reduced fetal weight and ossification; **repeated daily dosing** during organogenesis (extrapolated to the maximum human dose of **400 units**)

caused maternal toxicity, abortion, early delivery and maternal death, whereas **a single dose had no effect**.

- **BoNT-A only if benefit significantly outweighs risk; excretion in breast milk unknown** → caution in nursing mothers. A global safety database showed **fetal defect prevalence similar to background**, but very few urological indications were included.

FORMER LETTER CATEGORIES VERSUS PLLR

Letter categories (pre-2015)	PLLR (2015)
Single letter A, B, C, D, X	Narrative risk summary + data + clinical considerations
Quick classification	Requires individual interpretation of evidence
Pregnancy only	Pregnancy and lactation

AGENTS REFERRED TO ACROSS THE CHAPTER

Agent	Use	Status in pregnancy
Ampicillin, cephalexin (β -lactams)	Asymptomatic bacteriuria, UTI	Safe throughout
Nitrofurantoin	Bacteriuria, UTI if penicillin-allergic	Stop at 35 weeks — neonatal haemolytic anaemia
Morphine	Renal colic	Frequent small doses — traditional choice
Ketorolac / NSAIDs	Renal colic	Contraindicated — third-trimester ductus closure, fetal pulmonary hypertension
α -blockers, calcium channel blockers	Medical expulsive therapy	Off-label, inadequately studied — counsel
Unfractionated heparin	VTE prophylaxis	Low-dose in high risk; reversed by protamine
BoNT-A	Overactive bladder, neurogenic symptoms	Only if benefit clearly outweighs risk
Iodinated contrast	CT	Crosses placenta; no teratogenicity in animals
Gadolinium	MRI	Limit — only if clearly beneficial
^{99m}Tc -MAG3	Renography	Fetal dose <5 mGy — safe
BCG, mitomycin C	Intravesical therapy, non-muscle-invasive bladder cancer	Defer until after delivery
Multikinase inhibitors, mammalian target of rapamycin inhibitors, antiangiogenics, immunotherapy	Metastatic renal cell carcinoma	Avoid — human fetal risk
Corticosteroids	Fetal lung maturity	Give if viable preterm and delivery possible
Folic acid	Neural tube defect prevention	0.4 mg; 4–5 mg in myelomeningocele

- The drug table sorts by **what can be given, what must stop, and what waits for delivery**. **Traps:** NSAIDs for colic; nitrofurantoin at term; assuming contrast or radionuclide imaging is barred.

Concept — The PLLR replaced a label with a reasoning task: *each drug is judged on its evidence for this patient, not a letter.*

UROLITHIASIS IN PREGNANCY

- **Incidence 1:200 to 1:1500 pregnancies; renal colic is the leading nonobstetric cause of hospitalisation in pregnancy.**
- Stones associated with **preterm delivery, pre-eclampsia, placenta praevia, preterm premature rupture of membranes**, yet **no effect on birth weight, Apgar or perinatal mortality.**
- Historically **no higher than in age-matched nonpregnant women**; recent data: odds **rise with gestation, peak 0–3 months postpartum, baseline by 1 year.**
- Multiparous women reported to be affected more often (~3:1), but **not after adjustment for age**; population data link **prior pregnancy to later nephrolithiasis.**
- **Left and right calculi are equally common despite right-predominant hydronephrosis**; most present in the **second or third trimester.**

Etiology

- Recap: dilatation evident by **6–10 weeks**, up to 90% by the third trimester, may **persist 12 weeks postpartum**; **physiological hydronephrosis is the commonest cause of dilatation and mimics colic.**
- **Humoral:** progesterone relaxes ureteric smooth muscle, possibly reducing peristalsis. **Mechanical compression is probably primary — right predominance and rare dilatation below the pelvic brim**; mechanical mechanism predominates even with altered urinary anatomy.

LITHOGENIC VERSUS PROTECTIVE CHANGES OF PREGNANCY

Lithogenic	Protective
↑ Renal plasma flow and GFR → ↑ filtered sodium, calcium, urate → hypercalciuria, hyperuricosuria	↑ Urinary citrate
Parathyroid hormone suppressed ; placental 1,25-dihydroxycholecalciferol → ↑ intestinal calcium absorption	↑ Urinary magnesium
Urinary pH raised , most in second trimester	↑ Urine output
Urinary stasis from dilatation	(↑ glycosaminoglycans — see Myelomeningocele)

- These urinary changes are held responsible for **accelerated ureteric stent encrustation.**
- **Pregnancy stones are most commonly calcium phosphate** — the product of **alkaline urine plus hypercalciuria.**

Concept — *Pregnancy loads the urine with calcium and alkali, which predicts calcium phosphate stones and fast stent encrustation, but balances it with inhibitors and dilution, so the net stone rate barely changes.*

Evaluation

- **Signs masked, pain localisation altered**; many obstructing stones are **first misdiagnosed as appendicitis, diverticulitis or abruption.**
- **Commonest presentation: flank pain with macroscopic or microscopic haematuria**, sometimes UTI; **painless haematuria is unusual with an obstructing stone.**
- **Always send urine culture — baseline pyuria lowers its sensitivity for infection.**
- **Persistent infection with a urea-splitting organism → consider calculi.**
- Irritative voiding symptoms, chills, nausea and vomiting are nonspecific and shared with other intra-abdominal conditions.

Multidisciplinary Guidelines

- Multi-institutional panel (urology, anaesthesiology, obstetrics, diagnostic and interventional radiology) synthesised the literature into point-of-care flowcharts. Selected statements:
1. **Multidisciplinary approach** to suspected symptomatic nephrolithiasis.
 2. Initial evaluation: history, obstetric history, examination, **urinalysis with reflex culture, basic metabolic panel, complete blood count**.
 3. **Renal-bladder US is first-line imaging**.
 4. Diagnosis uncertain and clinical status changes → **offer second-line imaging**.
 5. Symptoms controllable → **trial of passage with hydration and analgesia**.
 6. **Septic obstructing stone** → **urgent decompression with ureteric stent**.
 7. **Exchange ureteric stents every 4 weeks** until definitive treatment.
 8. Conservative failure → **ureteroscopy with laser lithotripsy first-line** in noncomplex cases.
 9. **Third-trimester ureteroscopy may raise preterm labour but is not excluded**.
 10. **Neuraxial anaesthesia preferred** over general where appropriate.
 11. Discharge with stent, nephrostomy or residual stone → **arranged urology and obstetric follow-up**.
 - Narrative reviews reach concordant algorithms and stress that **extracorporeal shock wave lithotripsy (SWL) is contraindicated**.

Imaging

- **US is the standard initial study**, but has **low sensitivity, moderate specificity** for calculi (limitations tabulated).
- **Colour Doppler ureteric jets** are sensitive for obstruction in general, but **unpredictable in pregnancy** — use with caution.
- **Doppler resistive index (RI)** rises with acute obstruction; **RI is unchanged by pregnancy and by physiological hydronephrosis**, so it can detect acute obstruction here; **a change in RI is more accurate than absolute RI**.
- **Transvaginal US** images the distal ureter and detects **far more distal ureteric stones than transabdominal US**.
- Some women undergoing intervention for a radiographically “detected” stone prove to have **no stone**.
- **MRI: calcium not visualised** — **stone is a filling defect in high-signal urine**; small stones missed; distinguishes physiological from calculous hydronephrosis; **half-Fourier acquisition single-shot turbo spin-echo (HASTE)** improves utility.
- **Intravenous pyelography and nuclear renography**: limited, rarely used.
- **Noncontrast CT is the gold standard in nonpregnant patients (~99% sensitivity and specificity)**; per ACOG, **not to be withheld** if necessary.
- **Low-dose CT: high positive predictive value**, shows nonurological pathology; **second line**; ultra-low-dose protocols promising.
- **American Urological Association (AUA)**: radiologist and obstetrician estimate exposure so that **total pregnancy exposure does not exceed the ACOG maximum of 50 mGy**.

IMAGING FOR SUSPECTED UROLITHIASIS IN PREGNANCY

Modality	Role	Strength	Limitation
Renal-bladder US	First line	No radiation	Operator-dependent; cannot separate physiological from obstructive dilatation
	Adjunct	Sensitive in nonpregnant	Unpredictable in pregnancy

Modality	Role	Strength	Limitation
Colour Doppler jets			
Doppler RI	Adjunct	Unaffected by pregnancy ; Δ RI most accurate	Absolute RI less sensitive
Transvaginal US	Distal ureter	Detects far more distal stones	Distal segment only
MRI (HASTE)	Second line	No radiation; distinguishes cause of dilatation	Stone = filling defect; small stones missed
Low-dose CT	Second line	High positive predictive value; other pathology	Ionising (keep ≤ 50 mGy cumulative)
IVP, renography	Rarely	—	Fraught with limitations

Conservative management

- AUA: **spontaneous passage rates equal those of nonpregnant patients** → **observation is initial therapy if symptoms controlled**; 50–80% pass with hydration and analgesia. Follow closely, since a stone event raises maternal and fetal morbidity.
- Analgesia: traditional regimen is **frequent small doses of morphine**; coordinate with obstetrics.
- NSAIDs such as ketorolac are **contraindicated** — **premature closure of the ductus arteriosus and fetal pulmonary hypertension** if given in the third trimester.
- **Medical expulsive therapy (MET)** with α -blockers or calcium channel blockers relaxes ureteric smooth muscle; AUA endorses α -blockers for small uncomplicated distal stones in nonpregnant patients. In pregnancy: **inadequately investigated, off-label** — **counsel accordingly**.

Intervention

- **About one-third require intervention** — pain uncontrolled by analgesia, persistent obstruction, infection.
- **Delayed decompression raises maternal and fetal complications**; pregnant women with obstructing stones and UTI are **less likely to be decompressed** than nonpregnant women — a care gap.
- Options: **temporising (ureteric stent or PCN)** or **definitive (ureteroscopy, URS)**, planned by a multidisciplinary team.
- **Stents** drain well, but **hypercalciuria and hyperuricosuria accelerate encrustation** → frequent exchange (intervals tabulated below); early gestation therefore means **multiple exchanges**, with risks of **bacteriuria, UTI, migration, pain and reduced quality of life**.
- **PCN** shares the drawbacks — encrustation, colonisation, pain, **dislodgement**, repeated procedures — and is associated with **more preterm labour than observation, stenting or URS** (possibly confounded by acuity). Most tubes need exchange for **debris occlusion**; many are removed for obstruction, fever or pain.
- Temporising drainage usually still needs **postpartum definitive stone removal** → trend toward **definitive URS in any trimester**; complications uncommon and **urinary complications no different from nonpregnant patients**.
- **Radiation at URS**: **low-dose or pulsed fluoroscopy with collimation**, **lead shielding placed beneath the patient** (below-table source), **direct vision without fluoroscopy**, or **intraoperative US guidance**.
- **SWL**: **pregnancy is a contraindication**, despite inadvertent cases without fetal harm.
- **Percutaneous nephrolithotomy (PCNL)**: small series only; outside high-volume multidisciplinary centres (supine, US-guided), **defer until after delivery** (anaesthetic and radiation risk).

STENT VERSUS NEPHROSTOMY VERSUS URETEROSCOPY

	Ureteric stent	PCN	URS
Nature	Temporising	Temporising	Definitive
Indication	Septic obstruction; bridge	Alternative drainage	First-line after failed conservative care
Exchange	Every 4 weeks (guideline)	Every 3 weeks	—
Problems	Encrustation, UTI, migration, pain	Encrustation, dislodgement, occlusion, more preterm labour	Preterm labour in third trimester; radiation
Later need	Postpartum definitive treatment	Postpartum definitive treatment	Usually none

RECOMMENDED DRAINAGE EXCHANGE INTERVALS

Device	Interval	Source / setting
Ureteric stent	4–8 weeks	Stone comorbidity (review)
Ureteric stent	4–6 weeks	Several studies
Ureteric stent	4 weeks	Multidisciplinary guideline, known nephrolithiasis
Nephrostomy	3 weeks	Matched series (rapid encrustation, reinterventions)

KEY POINTS

- Ultrasonography remains the standard initial imaging study in evaluation of the pregnant patient.
- A trend toward offering definitive therapy with URS in pregnant patients has evolved with acceptable complication rates.
- Minimise ionising radiation during URS by using a below-table radiograph source and shielding the fetus with a lead apron placed below the patient.

Concept — *Diagnosis is ultrasound-led with low-dose CT a legitimate backstop; management escalates observation → temporising drainage → definitive ureteroscopy, and the burden of 3–4-weekly exchanges is what has pushed practice toward early definitive ureteroscopy.*

PREGNANCY IN WOMEN WITH CONGENITAL UROLOGIC CONDITIONS

- Systematic review of pregnancy after **neurogenic lower urinary tract dysfunction (NLUTD)** and **urological reconstruction** (1986–2020): **high risk of febrile UTI, premature delivery and delivery complications**; reassuringly, **obstetric outcomes are generally good with multidisciplinary care at a specialised centre**.
- Even with tertiary care, complications are frequent: **new or worsening hydronephrosis in most**, with many needing **nephrostomy**.
- Anorectal and cloacal malformations: **pregnancy feasible, with special risks**.

Myelomeningocele

- **Myelomeningocele (MMC)** considerations: age, comorbidity, **genetic risk to offspring**, pelvic bony and muscular anatomy, obesity, diversion status, prior abdominal surgery, **renal function**, fetal presentation → organise multidisciplinary care.

Fertility

- **Fertility normal; ~70% of those who conceive have successful pregnancies.**
- Uterus strains balance, **ventriculoperitoneal (VP) shunt**, bowel, bladder, diversion, skin, lungs, heart and — foremost — **UTI risk**.

- After extensive pelvic surgery, **adhesions → uterine retroflexion** → impaired conception and implantation.
- **Latex allergy up to 60%** → counsel on **latex-free condoms, sexual aids and intrauterine devices**.
- **Preconception counselling**; team of high-risk obstetrics, urology, anaesthesiology, orthopaedics, neurosurgery, plus social work.
- **Transmission if one parent affected: 1–8%, same whether mother or father**; incidence in female offspring **1 in 13**, male **1 in 50**; **both parents affected: 15%**.

Pregnancy

- Reconstruction preceded pregnancy by years and targeted kidneys and continence, not pregnancy.
- **Greatest risk factor for worsening renal function (with or without reconstruction) is pre-existing renal insufficiency**; pregnancy-related loss of function and new hypertension raise **preterm delivery and growth restriction**.
- Possible **solitary kidney**. Measure GFR rather than rely on a single creatinine — altered habitus and low muscle mass.
- **Stones after bladder augmentation: 9–50%** — incomplete emptying, chronic bacteriuria, foreign bodies, mucus, metabolic acidosis, **enteric hyperoxaluria**, diet.
- Uterus compresses the ureter **against the pelvic brim** → worse hydronephrosis and stasis. **After lower tract reconstruction, hydronephrosis worsens at 20–28 weeks — monitor most closely then**.
- **Inhibitors (citrate, magnesium, glycosaminoglycans)** mean even known stone formers do not form more stones in pregnancy, but **impaired sensation can hide upper tract obstruction until systemic illness**.
- **Retained defunctioned bladder** after diversion without cystectomy: **pyocystis may first become overt in pregnancy**.
- Other problems: compromised **mesenteric perfusion** of the diversion segment, intestinal obstruction, stomal stenosis or prolapse, **channel compression**, metabolic complications.
- **Ureterosigmoidostomy**: deformation of sigmoid or ureteric anastomoses → disrupted implants or incomplete emptying; **vaginal delivery can damage the anal sphincter** on which it depends → **de novo urinary ± faecal incontinence**.

Mode of Delivery

- **First obtain operative records, urodynamics and upper tract studies**; variable paediatric practice and revisions make the anatomy the priority.
- Determinants beyond obstetric factors: **type of diversion or prior surgery, disease anatomy, fetal presentation**.
- Native urethra or ureterosigmoidostomy → **continence may suffer after vaginal delivery**; counsel.
- **Augmentation with native bladder neck and urethral continence: vaginal delivery not contraindicated**. **Bladder neck reconstruction** risks injury to pelvic support and continence mechanism → **most recommend caesarean** (also with artificial urinary sphincter).
- **Artificial urinary sphincter (AUS)**: a functioning device **does not change pregnancy or delivery complications**; **avoid high forceps**; evidence anecdotal.
- **Prior bladder neck sling does not preclude vaginal delivery**; after synthetic **midurethral sling**, only a moderate fall in continence after either mode, most delivering vaginally. Continence loss after vaginal delivery following any incontinence procedure approaches **one in five** — discuss.
- **Vaginal: avoid prolonged labour** (pelvic floor). **Caesarean: urologist familiar with the anatomy** protects bladder and pedicle. Distance from high-risk care may favour **planned caesarean**.

- **Pelvic organ prolapse (POP)** common at baseline, worsened by pregnancy and either mode; **if POP needed repair before pregnancy, avoid vaginal delivery**. Limited **hip abduction** may preclude vaginal delivery.
- Depending on lesion level: **inadequate Valsalva for the second stage**; uterus may worsen **restrictive airway disease**.

DELIVERY CONSIDERATIONS IN MYELOMENINGOCELE

Vaginal delivery contraindicated	Vaginal delivery with caution	Concerns at caesarean section
Narrow bony pelvis	Ureterosigmoidostomy	Intraperitoneal VP shunt
Artificial sphincter or bladder neck reconstruction	Fetal malpresentation	Diversion pouch, enterocystoplasty, neobladder
Contracted hips	Uterine prolapse	

- The table separates **structural barriers to vaginal birth** from **continence-mechanism risk** and from **what is endangered at laparotomy**. **Trap**: augmentation alone with an intact bladder neck is **not** a contraindication to vaginal delivery.
- Caesarean technique: **high midline incision** to avoid the reservoir; **catheterise channels immediately preoperatively** to avoid or recognise injury.
- **VP shunt**: malfunction → headache, vomiting, impaired consciousness; obstruction from **raised intra-abdominal pressure** or mechanically; **infection at intra-abdominal surgery**, especially **emergency caesarean** with **spillage of contaminated urine**.
- Detecting **labour onset depends on lesion level**; paraplegia does not affect uterine contractility, but the **coordinated expulsive effort** of late labour may be deficient.

Prevention of Neural Tube Defects

- **Polymorphisms in folic acid metabolism** are considered the primary deficit → folate supplementation is critical.
- Early-1990s US programme (Centers for Disease Control and Prevention 1993, with Health Services Administration, FDA, National Institutes of Health), three stages: **(1) 0.4 mg folic acid for low-risk, 4 mg for high-risk women; (2) improved diet; (3) fortification of the food supply**.
- **Women with MMC: 4–5 mg daily** (versus 0.4 mg routine), **from 3 months before conception through 12 weeks of gestation**.
- Residual risk despite folate: inborn errors of metabolism, variable absorption; genetic, dietary and epigenetic factors under study.
- Prenatal screening: **maternal serum α -fetoprotein plus US; fetal MRI** as adjunct for prognosis.

Classic Bladder Exstrophy

- **Bladder exstrophy (BE): ~1 in 30,000 live births, male predominance**. Most now reach adulthood, but **adult continuity of care (pelvic floor, sexual function, fertility, pregnancy) is poor**.

Genital Anatomy

- **Pubic diastasis; anterior displacement of labia majora, vagina and anus; bifid clitoris**; bladder protruding through the abdominal wall defect.
- Bony and muscular defects compromise **levator ani anterior support**.
- **Vagina: reduced depth, normal calibre; introitus narrow, stenotic, anteriorly displaced; horizontal vaginal lie**. Diastasis widens with growth → divergent labia, mons and clitoris.

- **Deficient cardinal ligaments** → **poor apical support**; **cervix lies near the introitus on the anterior wall** → uterine prolapse.
- **Pelvic osteotomy at primary closure does not cause uterine prolapse** — the width of the adult diastasis is the determinant.
- **POP 18–30%**; the defining feature is **accelerated onset (mid-teens)**, **independent of parity**; many need surgery; higher still with continent diversion.

Fertility

- **Normal to mildly impaired**; **~68% of those who conceive have successful pregnancies**.
- Adhesions and retroflexion after pelvic surgery; **anterior-wall cervix may impair semen delivery**.

Pregnancy

- Preconception counselling and team as for MMC, plus continence.
- **Heritability ~1 in 70 offspring** — counsel.
- **POP in pregnancy managed conservatively** — bed rest and pessary; surgery deferred until childbearing complete. **Previously repaired POP tends to recur**; **acute postpartum POP in nearly half**; POP may hinder urethral self-catheterisation. Unlike the general population, **no parity-persistent POP association** detected.
- **Spontaneous abortion** high in some series (about one-third), at general-population rates in others.
- **Pre-eclampsia** in about a third in one series, mainly with **baseline renal insufficiency**.

Mode of Delivery

- Anatomy and postpartum prolapse → **planned caesarean with a multidisciplinary team is the accepted standard**.
- With any augmentation: **high midline or paramedian incision** to spare the **reservoir and augment mesentery**; catheterise channels preoperatively.
- Counsel on **preterm delivery** — **mean gestation ~36 weeks**; **abnormal presentation in over half** → extensive planning.
- **Emergency caesarean is far riskier than elective** → live near high-risk obstetric and urological expertise.

MYELOMENINGOCELE VERSUS BLADDER EXSTROPHY IN PREGNANCY

	Myelomeningocele	Bladder exstrophy
Fertility	Normal; ~70% successful	Normal–mildly impaired; ~68% successful
Transmission	1–8% (one parent); 15% both	~1 in 70
Dominant local issue	Renal function, VP shunt, latex allergy, sensation	POP — early, parity-independent
Mode of delivery	Individualised (see table above)	Planned caesarean
Caesarean incision	High midline	High midline or paramedian
Prevention	Folic acid 4–5 mg	—

KEY POINTS

- Most patients with congenital conditions affecting the lower urinary tract have normal fertility and should have prenatal counselling.
- Complexities of pregnancy after prior urinary diversion require an individualised approach.

- Mode of delivery depends on prior reconstruction and patient preference, although caesarean delivery is preferred in most patients with BE.

Concept — *Fertility is largely preserved; the risk lies in the reconstruction — infection, obstruction, prolapse and injury at delivery — so records, planning and a urologist at caesarean matter more than the mode of delivery itself.*

FUNCTIONAL AND ANATOMIC UROLOGIC CONCERNS DURING PREGNANCY

Intermittent Catheterization

- Uterine growth plus habitus and reduced flexibility make **intermittent catheterisation harder** — through a continent channel **and through the native urethra**; some women need an **indwelling catheter** for the pregnancy.

Urinary Diversion

- Heterogeneous population — **no global principles**, but series show **excellent outcomes**.

Ileocystoplasty (Bladder Augmentation)

- **Ileal conduit pedicle is displaced cephalad and lateral** to the uterus; **augmentation pedicle usually stays anterior to the uterus**.
- Tension on the mesentery rises slowly → **time to adapt**; **mesenteric adherence to the uterus is rare** but must be planned for.
- Literature is small case series, but **many augmented women have successful pregnancies and vaginal deliveries**.
- **High UTI rates and tendency to premature delivery**.
- Without bladder neck closure: **de novo incontinence** from uterine pressure, usually **resolving after delivery**.
- **Urine pregnancy tests give false positives after contact with bowel mucosa** → use serum human chorionic gonadotrophin (hCG).

Continent Catheterizable Stoma

- **De novo stomal difficulty**, especially a **right lower quadrant channel**; uterus compresses and stretches the channel; **resolves after delivery**.
- Options: **coudé-tipped catheter**; **native urethral catheterisation** if patent and feasible.
- Raised intra-abdominal pressure → **stomal prolapse or parastomal hernia**, likewise expected to resolve.

Neobladder

- Few reported cases after orthotopic neobladder; **caesarean section with urology in the team** was the preferred mode.

Urinary Tract Infection and Urinary Diversion

- **Asymptomatic bacteriuria (ASB): ≥ 1 bacterial species at $\geq 10^5$ colony-forming units (CFU)/mL ($\geq 10^8$ CFU/L), irrespective of pyuria, without signs or symptoms attributable to UTI.**
- Continent diversion or augmentation + intermittent catheterisation → **chronic or intermittent bacteriuria expected**; **outside pregnancy it is usually not treated**.
- **In pregnancy most experts treat ASB** because of fetal risk.
- **Pyelonephritis rises to 16–18% after urinary diversion; UTI/pyelonephritis up to 60% after augmentation** → preterm labour, fetal loss, small-for-gestational-age infants.

- Prophylaxis debate: many advise **daily antibiotics** (weigh teratogenicity); a reasonable middle course reserves it for the **highest risk** — **baseline or new hydronephrosis, multiple symptomatic infections, prepregnancy pyelonephritis**; others use **culture surveillance and treat only symptoms**. **No consensus**.

BACTERIURIA IN THE DIVERSION PATIENT: NONPREGNANT VERSUS PREGNANT

Nonpregnant	Pregnant
Chronic bacteriuria expected, usually not treated	Most treat ASB (fetal risk)
Treat symptomatic infection	Consider daily prophylaxis in highest-risk; no consensus

Spinal Cord Injury

- ACOG guidance (2020) on **spinal cord injury (SCI)**: **UTI and pyelonephritis are the commonest reasons for hospitalisation**, with little management literature.
- **Screen for ASB**; consider a **longer antibiotic course** given high risk.
- **Autonomic dysreflexia: bladder decompression** for urgent management of distension, and exclude other noxious stimuli.

Concept — *The gravid uterus acts on reconstructions mechanically — stretching channels and pedicles and obstructing catheterisation, largely reversibly — while bowel-in-urine makes bacteriuria chronic, so these women are managed as a high-infection-risk group.*

UROLOGIC MALIGNANCY DURING PREGNANCY

- **Malignancy complicates ~1 in 1000 pregnancies**; commonest is **breast (1 in 5000)**; **urological cancers ~13 per 1,000,000 pregnancies**.
- Balance pregnancy, maternal health and treatment delay → **individualised, multidisciplinary care**.
- Many conditions surface **incidentally through antenatal screening**; haematuria, flank pain and hypertension may be **attributed to pregnancy**.
- **Pregnancy and cancer are both highly antigenic states tolerated by the immune system**; pregnancy is thought **not to alter tumour biology or progression**, though data are lacking; little data in immunocompromised patients.

Renal Tumors

- **Renal cell carcinoma (RCC) is the commonest urological tumour in pregnancy**. Presentation as usual: pain, haematuria, hypertension, mass.
- Delayed diagnosis (low suspicion, reluctance to image, symptom overlap) → **over half >7 cm or advanced** (International Network on Cancer, Infertility and Pregnancy registry, INCIP).
- Known risks: **tobacco, von Hippel–Lindau disease; no causal role for pregnancy**; parity, hormones and subtype data inconsistent.
- Imaging principles as above (ACOG, American College of Radiology and American Institute of Ultrasound in Medicine, reaffirmed 2021); **AUA: contrast-enhanced CT or MRI** to characterise mass, kidney and viscera.
- **Iodinated contrast crosses the placenta but is not teratogenic or mutagenic in animals; gadolinium remains controversial** → **contrast CT preferred when enhancement is needed for staging**.
- **Renal mass biopsy** when haematological, metastatic, inflammatory or infectious lesion suspected. Young patient → **genetic counselling** for familial syndromes (AUA).
- Treatment as nonpregnant: **nephron-sparing or radical nephrectomy**, timed by clinical status and trimester, including **caesarean with simultaneous oncological surgery**; team defines fetal risk and advises when termination may be appropriate.

- **Laparoscopic surgery feasible with low morbidity in any trimester.**
- Systemic therapy — multikinase inhibitors, **mammalian target of rapamycin inhibitors**, antiangiogenics, immunotherapy — poorly characterised; **mostly avoided** (human fetal risk: abortion, malformation).

Angiomyolipoma

- **Angiomyolipoma (AML):** benign, **second commonest renal tumour**, usually asymptomatic; many are **not associated with tuberous sclerosis complex**.
- **Rupture → haematuria, flank pain, life-threatening retroperitoneal haemorrhage.** Pregnancy predisposes: **hormonal environment, vascular volume, raised abdominal pressure**. Ruptured tumours are large (~10 cm).
- **Haemodynamically stable → most reach term; unstable → substantial fetal death.**
- **Imaging (including serial imaging during observation) benefits outweigh fetal risk.**
- Outside urgent surgery, **embolisation is treatment of choice**, despite preterm labour risk from post-embolisation syndrome.

Bladder Tumors

- Exceedingly rare; **urothelial carcinoma commonest**; adenocarcinoma, squamous cell carcinoma (SCC) and benign leiomyoma reported.
- **Haematuria is misdiagnosed as vaginal bleeding in about a fifth → catheterised specimen** to establish the source. Urgency, frequency and nocturia are common in pregnancy and raise little suspicion.
- Most are **non-muscle-invasive bladder cancer (NMIBC)**. **SCC (schistosomiasis-associated):** aggressive, poor prognosis despite radical cystectomy.
- AUA NMIBC and muscle-invasive guidelines are **not always fully applicable** in pregnancy.
- **Haematuria without infection → standard evaluation: cystoscopy plus upper tract imaging**, primarily CT; **US diagnoses only about half.**
- **Transurethral resection is safe at any gestation**; staging imaging often required.
- **NMIBC: defer intravesical adjuvant therapy until after delivery (INCIP).** **Bacille Calmette-Guérin (BCG)** used once without fetal harm but carries systemic risk; **mitomycin C** likewise usually delayed.
- **Muscle-invasive disease: poor prognosis, no standard; neoadjuvant chemotherapy and radical cystectomy** timed to trimester, extent and stakeholder wishes.

Concept — *Treat as in the nonpregnant patient — definitive surgery for renal tumours, resection for bladder tumours — but defer what can wait (intravesical and systemic therapy) and time what cannot to the trimester.*

MANAGEMENT OF URINARY TRACT INFECTIONS IN PREGNANCY

- **Up to 26% of women** report UTI diagnosis or treatment in pregnancy, with maternal and neonatal risk.
- **ASB in 2–10% of pregnancies** (definition above).
- Moderate-quality evidence: **treating ASB likely reduces progression to pyelonephritis.**
- **Untreated ASB → pyelonephritis 15–45%**; most episodes in the **third trimester**, when hydronephrosis and stasis peak; **one-third have prior pyelonephritis.**
- Pyelonephritis → **preterm birth** (ascending infection may underlie up to half of premature deliveries, especially <30 weeks), **septicaemia, respiratory distress, low birth weight.**
- **Treatment lowers pyelonephritis to 1–4%** and reduces preterm birth and low birth weight.
- **Universal screening** (US Preventive Services Task Force, USPSTF; Infectious Diseases Society of America, IDSA): **urine culture at 12–16 weeks or first antenatal visit**; positive = **midstream clean-catch >100,000**

CFU/mL of a single uropathogen. Insufficient evidence for repeat screening after a negative culture or after treatment.

- Organisms as in nonpregnant women; choose the agent with obstetrics.
- **β-Lactams (ampicillin, cephalexin) safe throughout; nitrofurantoin if penicillin-allergic, stopped at 35 weeks** (neonatal haemolytic anaemia).
- Duration: single dose showed a trend to **less clearance** than short courses → **IDSA 4–7 days**, shortest effective course.

Acute pyelonephritis

Admit → **parenteral antibiotics** (culture, local antibiogram, shared decision) until clinical improvement → tailor to sensitivities → **image if obstruction suspected or improvement not rapid** → **oral culture-directed agent to complete ≥14 days** → repeat cultures, nonpharmacological measures or **low-dose prophylaxis** in selected cases.

ASYMPTOMATIC BACTERIURIA: NONPREGNANT VERSUS PREGNANT

Nonpregnant	Pregnant
No screening	Screen all — culture at 12–16 weeks
Not treated	Treat — 4–7 days
Low progression	15–45% pyelonephritis untreated; 1–4% treated

Concept — *Stasis in the dilated tract lets bacteriuria ascend, which converts ASB from benign to dangerous — so pregnancy is the setting where ASB is screened for and treated.*

UROLOGIC MANAGEMENT OF PLACENTAL ABNORMALITIES

- **Disorders of invasive placentation** (placenta accreta spectrum): trophoblast invades the myometrium; “placenta accreta” is also used for the whole spectrum.

PLACENTA ACCRETA SPECTRUM

Grade	Depth	Urological relevance
Accreta	Attaches to myometrium, no invasion	Haemorrhage at separation
Increta	Villi invade myometrium	—
Percreta	Through full myometrium and serosa	Adheres to bladder, ureters, rectum — the form urologists meet

- Failed separation or invasion → **life-threatening haemorrhage and coagulopathy; average blood loss 3000–5000 mL; maternal mortality up to 7%** despite optimal planning.
- **Prior caesarean delivery is the predominant risk factor** (implantation in a poorly vascularised scar); **maternal age >35** contributes.
- **US incidence ~1 in 533 deliveries.**
- **Haematuria in ~25% of percreta.** With risk factors, **predelivery cystoscopy** assesses the posterior bladder wall — **do not biopsy the hypervascular tissue.**
- **Grey-scale US is the standard diagnostic modality; MRI when US is inconclusive or to map invasion** of vessels or adjacent organs (ureter).
- **Multidisciplinary team at a tertiary centre of excellence:** anaesthesia, obstetrics, gynaecological oncology, intensive care, maternal–fetal medicine, interventional radiology, urology, general surgery.
- **Planned preterm caesarean hysterectomy** — caesarean delivery, **placenta left in situ**, hysterectomy — **at ~34 weeks** in stable patients, with **antenatal corticosteroids**; individualise; have **emergency contingency plans.**

- **Unsuspected accreta with attempted removal of adherent villi → major morbidity.**
- **Conservative (fertility-sparing) management** — placenta left in situ ± uterine artery embolisation, methotrexate, haemostatic sutures, devascularisation, balloon tamponade: **delayed hysterectomy is the commonest outcome**; delayed bleeding and infection; role uncertain.
- Urological counselling: **ureteric catheters or stents**, possible **major injury, extirpative surgery or diversion**. Genitourinary injury: **~29% historically, ~20% in contemporary tertiary multidisciplinary series**.
- **Retrograde cystoscopic ureteric catheters** aid ureteric identification and recognition of injury (protective effect mixed). If hysterectomy is delayed, **convert to indwelling double-J stents**. **Up to one-third need re-exploration** after peripartum hysterectomy — stents help.
- **Attempt bladder preservation**; with known detrusor invasion, **planned cystotomy and partial cystectomy** rather than trying to develop a placenta-bladder plane (haemorrhage).
- Counsel on long-term **fistula, incontinence, neurogenic lower urinary tract dysfunction**.

KEY POINTS: MANAGEMENT OF PLACENTAL ABNORMALITIES

- US is superior to MRI for routine screening for placenta accreta in at-risk patients, but invasion of adjacent pelvic tissues and organs may be better defined by MRI.
- Optimal timing for planned delivery with suspected placenta percreta is approximately 34 weeks.
- Antenatally diagnosed percreta → tertiary centre and a team including urologists, general surgeons, obstetric anaesthesiologists, interventional radiologists and intensivists.
- Counsel carefully on complexity and urinary tract injury; consider retrograde catheter placement to prevent or identify ureteric injury.

Concept — *Percreta is a planned operation, not an emergency: diagnose early, deliver electively at ~34 weeks in a tertiary centre, protect the ureters with catheters, and resect invaded bladder rather than peel it.*